

Many broker networks have been developed with the high frequency trader in mind and are well equipped to handle large amounts of data, messages, and volume. The infrastructure is built on a flexible ultra-low latency FIX platform. Some of these brokers also provide smart order routing access, as they are often better prepared to monitor and evaluate level II data, order book queues, and trading flows and executions by venue in real-time.

Advantages

- *Lower Commissions.* Brokers are paid a fee by the fund to compensate them for their infrastructure and connectivity to exchanges, trading venues, dark pools, etc. This fee is usually lower than the standard commission fee and the fund does not receive any additional benefit from the broker such as order management services, risk management controls, etc.
- *Anonymity.* Orders are entered into the system and managed by the trader. Brokers do not see or have access to the orders.
- *Control.* Traders have full control over the order. Traders determine the venues (displayed/dark), order submission rules such as market/limit prices, share quantities, wait and refresh times, as well as when to accelerate or decelerate trading based on the investment objective of the fund and actual market conditions. Information leakage is minimized since the broker does not receive any information about the order or trading intentions of the investor.
- *Access.* Access to the markets via the broker's technology and infrastructure. This includes co-location, low latency connections, etc.
- *Perfectly Customized Strategies.* Since the investor defines the exact algorithmic trading rules, they are positioned to ensure the strategy is exactly consistent with their underlying investment and alpha expectations. Funds rarely (if ever) provide brokers with proprietary alpha estimates.

Disadvantages

- *Increased Work.* Funds need to continuously test and evaluate their algorithms, write and rewrite codes, develop their own limit order models and smart order routers.
- *Lack of Economies of Scale.* Most funds do not have access to the large number and breadth of orders entered by all customers. Therefore, they do not have as large a data sample to test new and alternative algorithms. Brokers can invest substantial resources in an algorithmic